Compendium of Bioagents in Agriculture:
Information on Biocontrol agents used in Biological control of crop insect pests

S.N	Chemical name	Common name	Concept of active ingredient in the formulation	Target organism /host	Physiological stage for application	Recommended dose & mode of application	Reason for recommendation	Side effects, if any	Not for use on
		Acerophagus papayae	*Endoparasitoids of papaya mealybug*	*On papaya mealybug Paracoccus marginatus*	Field release	250 per ha	To reduce mealybug infestation	Nil	
		Anagyrus loecki	*Endoparasitoids of papaya mealybug*	*On papaya mealybug Paracoccus marginatus*	Field release	250 per ha	To reduce mealybug infestation	Nil	
		Pseudleptomastyx mexicana	*Endoparasitoids of papaya mealybug*	*On papaya mealybug Paracoccus marginatus*	Field release	250 per ha	To reduce mealybug infestation	Nil	
		*Trichogramma chilonis*Ishii (I) [#] Hymenoptera: Trichogrammatidae	Parasitised egg cards	Sugarcane borers *Chilo infuscatellus, Chilo sacchariphagus indicus, Chilo auricilius, Acigona steniellus;* Cotton (Non Bt) bollworms *Helicoverpa armigera, Pectinophora gossypiella* & *Earias* spp.; Maize stem borer *Chilo partellus,* Diamond back moth *Plutella*	Field release	50,000/ha on sugarcane and vegetables; 100,000/ha on maize and 1,50,000/ha on cotton	**Sugarcane**: 4 to 6 releases at 10 days intervals for early shoot borer; 8 to 10 releases for stalk, internode and Gurdaspur borers **Cotton (Non Bt) & Vegetables:** Six weekly releases **Maize: Three** releases at five		

				xylostella; Tomato fruit borer *Helicoverpa armigera*			days intervals		
		Trichogramma japonicum (I) Hymenoptera: Trichogrammatidae	Parasitised egg cards	Top shoot borer of sugarcane *Scirpophaga excerptalis* and Paddy stem borer *Scirpophaga incertulas*	Field release	Sugarcane & Paddy: 50,000/ha	**Sugarcane:** 4 to 6 releases at 10 days intervals on observing pest or from 60th day **Paddy:** 6 releases on appearance of pest or from 30th day after transplantation		
		Trichogramma achaeae (I)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Cotton (Non Bt) bollworms and Bhendi Borer	Field release	1,50,000/ha on cotton (Non Bt) 50,000/ha on vegetables	Six releases at weekly intervals		
		Trichogramma pretiosum (E)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Tomato fruit borer *Helicoverpa armigera*	Field release	50,000/ha	Six releases at weekly intervals on appearance of pest or from 45th day from transplatation		
		Trichogramma embryophagum (E)* Hymenoptera:	Parasitised egg cards	Apple Codling moth *Cydia pomonella*	Field release	2000 adults per tree or 100,000/ha	Releases starting from the first moth catch, continue at weekly intervals till pest egg availability in the		

		Trichogrammatidae					field		
		Trichogramma dendrolimi Matsumara (E)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Targeted against tissue borers on maize and sugarcane – For Research work	Field release	-	-		
		Trichogramma brassicae (E)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Diamondback moth *Plutella xylostella* and Cabbage butterfly *Pieris brassicae* on cabbage and cauliflower	Field release	100,000/ha	Six releases at weekly intervals		
		Trichogramma evanescens Westwood Hymenoptera: Trichogrammatidae	Parasitised egg cards	Targeted against tissue borers on maize and sugarcane - For Research work	Field release	-	-		
		Trichogramma mwanzai (E)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Targeted against *Helicoverpa armigera* - For Research work	Field release	-	-		
		Trichogrammatoidea armigera (E)* Hymenoptera:	Parasitised egg cards	Targeted against *Helicoverpa armigera* - For	Field release	-	-		

		Trichogrammatidae		Research work					
		Trichogrammatoidea bactrae (E)* Hymenoptera: Trichogrammatidae	Parasitised egg cards	Diamond back moth *Plutella xylostella* on cabbage	Field release	2,50,000/ha	Five releases at weekly intervals		
		Telenomus remus Nixon (E) Hymenoptera: Scelionidae	Parasitised egg cards	Tobacco caterpillar *Spodoptera litura*	Field release	1 lakh /ha	Three to four releases		
		Goniozus nephantidis (Muesebeck) (I) Hymenoptera: Bethylidae	Cocoons	Coconut black-headed caterpillar *Opisina arenosella*	Field release on tree trunks	10 adults per palm	Four releases		
		Chelonus blackburnii Cameron (E)* Hymenoptera: Braconidae	Adults	Potato tuber moth *Phthorimaea operculella*	Field release	50000 adults /ha in the field 2 adults per kg of potatoes in godowns	Two releases at weekly intervals Three to four releases (or as per need) at fortnightly intervals		
		Cryptolaemus	Adults / Grubs	Mealy bugs	Field release	10 beetles or 50	One or more		

		montrouzieri Mulsant (E) Coleoptera: Coccinellidae		*Maconellicoccus hirsutus, Planococcus citri, P. lilacinus*		grubs /infested plant or tree or 5000 beetles/ha	releases based on pest intensity		
		Scymnus coccivora (Ramakrishna Ayyar) (I) *Coleoptera: Coccinellidae*	Adults	Mealy bugs on citrus, grapes and other fruit crops (*M. hirsutus, Planococcus* spp.	Field release	600 – 2500 adults/ha	One or more releases based on pest intensity	Nil	
		Chilocorus nigrita (Fabricius) (I) Coleoptera: Coccinellidae	Adults / Eggs	Sugarcane scale insect *Melanaspis glomerata* Citrus scale *Aonidiella aurantii*	Field release	1500 beetles/ha; or 10 egg pads (with 40 eggs per pad) in 100 spots/ha (40,000 eggs/ha) 10 adults/tree	One or more releases based on pest intensity		
		Cheilomenes sexmaculata Fabricius (I)* Coleoptera: Coccinellidae	Adults / Eggs	*Aphis craccivora* on legumes and *Lipaphis erysimi* on oilseed crops	Field release	5000 larvae or 500 adults per ha	Two releases; first release to coincide with the appearance of aphids		
		Coccinella septempunctata Linnaeus (I)* Coleoptera:	Adults / Eggs	*Aphis craccivora* on legumes and *Lipaphis erysimi* on	Field release	5000 larvae or 500 adults per ha	Two releases; first release to coincide with the appearance of		

		Coccinellidae		oilseed crops			aphids		
		Brumoides suturalis (Fabricius) (I)* Coleoptera: Coccinellidae	Adults	Aphids and white flies	Field release	-	-		
		Curinus coeruleus Mulsant (E) Coleoptera: Coccinellidae	Adults	Subabul psyllid *Hetropsylla cubana*	Field release	20 Adults per tree	Two releases during July and October		
		Chrysoperla carnea (Stephens) (I) Coleoptera: Coccinellidae	Eggs / First instar larvae	Sucking pests on cotton, tobacco, sunflower, groundnut& some fruit crops	Field release	10,000 first instar larvae/ha	Twice during the season with an interval of 15 days On fruit crops, 10 – 20 larvae per infested tree		
		Mallada spp. (I) * Neuroptera: Chrysopidae	Cocoons	Sucking pests on cotton, tobacco, sunflower, groundnut& some fruit crops	Field release	10,000 first instar larvae/ha	Twice during the season with an interval of 15 days On fruit crops, 10 – 20 larvae per infested tree		
		Ischiodon scutellaris (Fabricius) (I) * Diptera: Syrphidae	Cocoons	*Aphis craccivora* on legumes and *Lipaphis erysimi* on oilseed crops	Field release	5000 larvae/ha	-		
		Micromus timidus (Hagen) (I) *	Larvae	*A. craccivora* on legumes & oilseeds	Field release	Research in	Research in		

		Neuroptera: Hemerobidae		& Sugarcane Woolly Aphid		progress	progress		
		Cardiastethus exiguus Poppius (I) Hemiptera: Anthocoridae	Adults/Nymphs	*Opisina arenosella* Coconut black-headed caterpillar		50 nymphs/adults per tree	Three releases		
		Blaptostethus pallescens Poppius (I) Hemiptera: Anthocoridae	Adults/Nymphs	Spider mites on bhendi	Field release	5 to 10 nymphs per plant	Five releases		
		Xylocoris flavipes (Reuter) (I) * Hemiptera: Anthocoridae	Adults/Nymphs	Storage pests	Field release	For research work	-		
		Orius tantillus / Orius maxidentex. (I) Hemiptera: Anthocoridae *	Adults/Nymphs	Thrips	Field release	For research work	-		
		Heterorhabditis indica (Entomo pathogenic nematodes)	Infective juveniles in wettable powder	White grubs, Root weevils & soil pests	Soil Application	20kg/ha	For combating Soil insect pests and reducing chemical insecticide usage		
		Bacillus thuringiensis	*Bt* crystals and Bt spores 5%	Lepidopteran, coleopteran and dipteran pests of crops	Foliar spray	1kg/ha	To reduce the use of chemical pesticides	Nil	-
		Spodoptera litura Nuclear Polyhdrosis	Poly hedral bodies of the	*Spodoptera litura*	Foliar spray	250 Larval Equivalent (LE)	To reduce the use of		

		virus(Sl NPV)	NPV Virus 1 X 10^9PIB/ml			/ha	chemical pesticides		
		Helicoverpa armigera Nuclear Polyhdrosis virus(Ha NPV)	Poly hedral bodies of the NPV Virus 1 X 10^9PIB/ml	*Helicoverpa armigera*	Foliar spray	250 Larval Equivalent (LE) /ha	To reduce the use of chemical pesticides		
5		*Beauveria bassiana*	Spore cum mycelia formulation 1X10^8 CFU/g/ml	Several insect pests of crops	Foliar spray for foliar pests Soil application for soil pests	2.5-5.0kg/ hafor foliar spray 2.5-5.0 kg+250-500kg FYM /ha for soil application	For combating insect pests and reducing chemical insecticide usage	-Nil-	-
		Metarhizium anisopliae	-do-	-do-	-do-	-do-	-do-	-Nil-	-
		Verticillium lecanii	-do-	Sucking pests of various crops	Foliar spray	2.5-5.0kg/ hafor foliar spray	-do-	-Nil-	-
		Paecilomyces fumosoroseus	-do-	For mite control	-do-	2.5-5.0kg/ hafor foliar spray	-do-	-Nil-	-

Compendium of Bioagents in Agriculture:
Information on Biocontrol agents used in Biological control of nematodes

S.N	Chemical name	Common name	Concept of active ingredient in the formulation	Target organism /host	Physiological stage for application	Recommended dose & mode of application	Reason for recommendation	Side effects, if any	Not for use on
1		*Pochonia chlamydosporia*	Spore cum mycelia formulation	Plant parasitic nematodes (PPN)	Soil application	15kg/ha	For combating PPN & reducing chemical nematicide usage	-Nil-	-
2		*Paecilomyces lilacinus*	Spore cum mycelia formulation	-do-	-do-	-do-	-do-	-Nil-	-

Compendium of Bioagents in Agriculture:
Information on Biocontrol agents used in Biological control of crop diseases

S.N	Chemical name	Common name	Concept of active ingredient in the formulation	Target organism /host	Physiological stage for application	Recommended dose & mode of application	Reason for recommendation	Side effects, if any	Not for use on
1		*Trichoderma viride*	Spore cum mycelia formulation of 2 X 10^6 CFU/g/ml	Soil-borne plant pathogens like *Rhizoctonia, Phythium, Phytophthora, Scleroterium, Fusarium* etc.	Seed treatment and soil application	Seed treatment dosage: 4-10g/kg seeds Soil application dosage: 2.5-5.0 kg+250-500kg FYM /ha for soil application	For combating soil borne plant pathogens and reducing chemical fungicide usage	-Nil-	-
2		*T. harzianum*	-do-	-do-	-do-	-do-	-do-	-Nil-	-
3		*T. virens*	-do-	-do-	-do-	-do-	-do-	-Nil-	-
4		*Pseudomonas fluorescens*	The formulation consists of live cells 1×10^8 CFU/g/ml	Fungal pathogens	Seed treatment Foliar spray	Seed treatment - 10g per Kg of seeds. Foliar spray- 0.1% of formulation mixed in water	Suppression of plant pathogens. Plant growth promoter.	Nil	
5		*Bacillus subtillis*	The formulation consists of live cells 1×10^8 CFU/g/ml	Fungal pathogens	Seed treatment Foliar spray	Seed treatment - 10g per Kg of seeds. Foliar spray- 0.1% of formulation mixed in water	Suppression of plant pathogens. Plant growth promoter.	Nil	